

**Started on** Monday, 24 August 2026, 5:20 PM**State** Finished**Completed on** Monday, 24 August 2026, 5:36 PM**Time taken** 15 mins 55 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
        printf("*");
    }
    else
    {
        for(int i=1; i<=n; i++)
        {
            for(int j=1; j<=n; j++)
            {
                printf("*");
                printf("*");
                break;
            }
        }
    }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

```
1  #include <stdio.h>
2
3  int main() {
4      int n;
5      scanf("%d", &n);
6
7      int count = 0;
8
9      if (n == 1) {
10         count++;
11     } else {
12         int i = 1;
13         count++; // 1: initialization of i = 1
14
15         while (i <= n) {
16             count++; // 1: i <= n check (true)
17
18             int j = 1;
19             count++; // 2: j = 1 initialization
20
21             while (j <= n) {
22                 count++; // 3: j <= n check (true)
23                 count++; // 4: inner loop body execution
24                 break; // exits inner while loop only
25             }
26
27             i++;
28             count++; // 5: i++ increment
29         }
30         count++; // 2: final i <= n check (false)
31     }
32
33     printf("%d\n", count);
34
35     return 0;
}
```

36 | }

	Input	Expected	Got	
✓	2	12	12	✓
✓	1000	5002	5002	✓
✓	143	717	717	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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